

Practice Questions on GHCR

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Q #1:

You are required to build a Docker image from a Dockerfile and publish it to GitHub Container Registry (GHCR), and then pull it on another system.

Task Requirements:

1. Create a simple Dockerfile for a Python Flask application.
2. Build the Docker image locally and tag it appropriately for GitHub Container Registry (ghcr.io).
3. Authenticate Docker with GitHub Container Registry using a Personal Access Token (PAT).
4. Push the Docker image to ghcr.io/<your-github-username>/<image-name>:v1.
5. Verify that the image is successfully uploaded in your GitHub Packages section.
6. On a different system (or after removing the local image):
 - Login again to GHCR.
 - Pull the same image.
 - Run the container and verify the application is working.

Q#2: You are required to create a customized MySQL Docker image with a preloaded database and publish it to GitHub Container Registry (GHCR), then pull and verify it on another system.

Task Requirements:

1. Create a custom Dockerfile based on the official MySQL image.
2. Configure the Dockerfile to:
 - Set root password using environment variables.
 - Create a new database (e.g., college_db).
 - Import a sample SQL file (init.sql) containing:
 - One table (e.g., students)
 - At least 3 records.
3. Build the Docker image locally.
4. Tag the image properly for GitHub Container Registry:

Evaluation Rubrics:

- Correct use of MySQL base image
- Proper database initialization
- Correct GHCR tagging format
- Successful push and pull
- Verification of database and table after pulling